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PHYSICS OF FLUIDS **24**, 035110 (2012)

Rare backflow and extreme wall-normal velocity fluctuations in near-wall turbulence

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Rare negative streamwise velocities and extreme wall-normal velocity fluctuations near the wall are investigated for turbulent channel flow at a series of Reynolds numbers based on friction velocity up to $Re_\tau = 1000$. Probability density functions of the wall-shear stress and velocity components are presented as well as joint probability density functions of the velocity components and the pressure. Backflow occurs more often (0.06% at the wall at $Re_\tau = 1000$) and further away (up to $y^+ = 8.5$) from the wall for increasing Reynolds number. The regions of backflow are circular with an average diameter, based on ensemble averages, of approximately 20 viscous units independent of Reynolds number. A strong oblique vortex outside the viscous sublayer is found to cause this backflow. Extreme wall-normal velocity events occur also more often for increasing Reynolds number. These extreme fluctuations cause high flatness values near the wall ($F(v) = 43$ at $Re_\tau = 1000$). Positive and negative velocity spikes appear in pairs, located on the two edges of a strong streamwise vortex as documented by Xu *et al.* [Phys. Fluids **8**, 1938 (1996)] for $Re_\tau = 180$. The spikes are elliptical and orientated in streamwise direction with a typical length of 25 and a typical width of 7.5 viscous units at $y^+ \approx 1$. The negative spike occurs in a high-speed streak indicating a sweeping motion, while the positive spike is located in between a high and low-speed streak. The joint probability density functions of negative streamwise and extreme wall-normal velocity events show that these events are largely uncorrelated. The majority of both type of events can be found lying underneath a large-scale structure in the outer region with positive sign, which can be understood by considering the more intense velocity fluctuations due to amplitude modulation of the inner layer by the outer layer. Simulations performed at different resolutions give only minor differences. Results from experiments and recent turbulent boundary layer simulations show similar results indicating that these rare events are universal for wall-bounded flows. In order to detect these rare events in experiments, measurement techniques have to be specifically tuned.
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I. INTRODUCTION

Turbulent channel flow serves as a model for wall-bounded turbulence and direct numerical simulations (DNSs) of such flows have improved our understanding of the phenomena occurring in wall-bounded turbulent flows. The behaviour of the flow in the near-wall region has a large influence on the drag acting on an immersed body and thus it is of great importance to completely comprehend the physics of near-wall turbulence, both from a fundamental and an engineering point of view.

The occurrence of negative streamwise velocities in wall-bounded turbulence is a counter-intuitive phenomenon. Eckelmann[1] stated in 1974 that "with certainty, there are no negative velocities near the wall." More recent, Colella and Keith[2] concluded that "The probability density of wall-shear

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stress fluctuations (...) exhibits no flow reversals at the wall." However, because of the presence of pressure in the Navier-Stokes equations, the possibility of negative streamwise velocities cannot be excluded. This is not the case for a scalar transport problem, e.g., in studies concerning a passive scalar, the scalar cannot exceed the limiting values set by the boundary conditions. Indeed, Spalart and Coleman[3] reported negative wall-shear stress for a zero-pressure gradient (ZPG) turbulent boundary layer (TBL) simulation, and Hu *et al.*[4] discussed negative wall-shear stress in a turbulent channel. Similar results were determined by laser Doppler velocimetry (LDV) measurements by Johansson[5] in a ZPG TBL. In the experiments clear instantaneous flow reversal has been observed and it is explicitly noted that this is not due to measurement noise. The author concluded that "The frequency of occurrence and the importance of these samples, though, need to be investigated in more detail," but to the best of our knowledge no further research on this topic has been performed. Furthermore, Chong *et al.*[6] only mentioned "local three-dimensional separated flow at the surface, caused by local vorticity and pressure gradients" in a turbulent channel flow.

Another rare event in the near-wall region of wall-bounded flows is large fluctuations of the wall-normal velocity component, up to an order of magnitude larger than their local standard deviation. This causes extreme flatness values of the wall-normal velocity which have been reported by several previous studies, see, e.g., Lyons *et al.*,[7] Xu *et al.*,[8] Manna and Vacca,[9] and Durst and Beronov.[10] Since the high values were initially only observed in direct numerical simulations but not in experiments, it was thought that these values were non-physical, and in fact numerical artifacts,[10] or that they were caused by the recycling of strong disturbances due to the use of periodic boundary conditions.[7] However, by specifically designed LDV measurements, Xu *et al.*[8] reported that similar high flatness levels are also found in experiments. They further attributed the high flatness values to strong sweep events which only happen very close to the wall and are rare in both time and space. Manna and Vacca[9] also suggested that these values are caused by rare but energetic events near the wall.

Although flow reversal is a counterintuitive phenomenon, its existence coupled with the occurrence of extreme wall-normal fluctuations brings along that the practice guides of current measurement techniques may need to be revised in order to allow the detection of instantaneous reverse flow as well as to capture these rare and extreme events.

To investigate these rare events and to try to find whether they are related, we performed and analysed DNS of turbulent channel flow up to Reynolds number of $Re_\tau = 1000$ based on the friction velocity u_τ and channel halfwidth h. This substantially expands the Reynolds number range of Xu *et al.*[8] who considered DNS at $Re_\tau = 180$. Because these events occur seldom, long integration times are required to obtain statistically relevant data. An extensive study was carried out to ensure convergence of the statistics of interest. Probability density functions (PDFs) of wall-shear stress and velocities, and joint PDFs (JPDF's) of velocity components and pressure are computed and compared. In addition, possible Reynolds number dependencies and trends as well as resolution effects are studied. Visualisations of instantaneous flow fields and ensemble averages are used to interpret the physical phenomena causing these events.

II. DNS OF INCOMPRESSIBLE TURBULENT CHANNEL FLOW

Fully developed plane periodic channel flow is used as the flow case to investigate these rare events. The channel halfwidth h is employed as the reference length and the calculated friction velocity u_τ as the reference velocity which leads to a Reynolds number $Re_\tau = u_\tau h/\nu$ with ν being the kinematic viscosity. The DNSs were performed with *SIMSON*, a fully spectral in-house code described by Chevalier *et al.*[11] It uses Fourier expansions in the horizontal directions and Chebyshev polynomials in the wall-normal direction to discretise the governing equations and boundary conditions. Time is advanced using a mixed Runge-Kutta/Crank-Nicolson scheme, while the non-linear terms are calculated in physical space with aliasing errors removed by the 3/2-rule. After reaching a statistically steady state, the statistics are averaged over a certain time period T. In this time period, a certain number of instantaneous velocity fields N_f is stored for each simulation.

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TABLE I. Computational box and resolution of the different simulations. L_x, L_y, and L_z are the length in streamwise (x), wall-normal (y), and spanwise (z) direction; N_x, N_y, and N_z are the corresponding number of points in physical space; Δx^+, Δy^+_{max}, and Δz^+ are the grid spacing in viscous units; N_f is the number of instantaneous velocity fields used; and $u_\tau T/h$ is the number of turnover periods for eddies of size h and velocity u_τ.

Re_τ nom.	Re_τ calc.	$(L_x \times L_y \times L_z)/h$	$N_x \times N_y \times N_z$	Δx^+, Δy^+_{max}, Δz^+	N_f	$u_\tau T/h$
180*c*	178.8	$4\pi \times 2 \times \frac{4}{3}\pi$	$128 \times 129 \times 128$	18, 4, 6	301	128
180*m*	178.9	$4\pi \times 2 \times \frac{4}{3}\pi$	$192 \times 129 \times 160$	12, 4, 5	261	111
590*m*	586.4	$2\pi \times 2 \times \pi$	$384 \times 257 \times 384$	10, 7, 5	43	30
590*f*	587.1	$2\pi \times 2 \times \pi$	$768 \times 385 \times 768$	5, 5, 2.5	49	34
1000*m*	1000.2	$8\pi \times 2 \times 3\pi$	$2560 \times 385 \times 1920$	10, 8, 5	2	13

Simulations at $Re_\tau = 180$, 590, and 1000 were performed. The computational domain and resolutions for the various simulations are summarised in Table I. Both the simulations at $Re_\tau = 180$ and 590 were performed with two different resolutions to investigate potential resolution effects. To distinguish between the coarse, medium, and fine resolutions, a "*c*," "*m*," or "*f*" is appended to their name, each letter corresponding to similar grid spacing in viscous units. The coarse grid has a spacing of about $(\Delta x^+, \Delta y^+_{max}, \Delta z^+) = (18, 4, 6)$, the medium grid has (10, 7, 5), and the fine grid has (5, 5, 2.5).

Figure 1 shows that the present channel results agree very well with previous DNS data by Moser *et al.*[12] and del Álamo *et al.*[13] Both the mean and root mean square (rms) ($u_{\rm rms} = \langle (u - \langle u \rangle)^2 \rangle^{1/2} = \langle u'^2 \rangle^{1/2}$) of the streamwise velocity coincide with previous data. The mean streamwise velocity in viscous scaling collapses well in the near-wall region, while the rms velocity increases for increasing Reynolds number, with the maximum occurring around $y^+ = 15$, in accordance with previous channel and ZPG TBL simulations (see, e.g., Schlatter and Örlü[14]). Note that the differences between different resolutions are small, such that they are almost indistinguishable in Figure 1. Furthermore, to ensure that also higher order moments are converged, we performed a convergence study from which we conclude that the present data has converged to a satisfactory statistically steady state. After reaching this state, all moments up to the fourth order vary less than ±2.5% around their respective mean values. More information on this convergence study can be found in the Appendix.

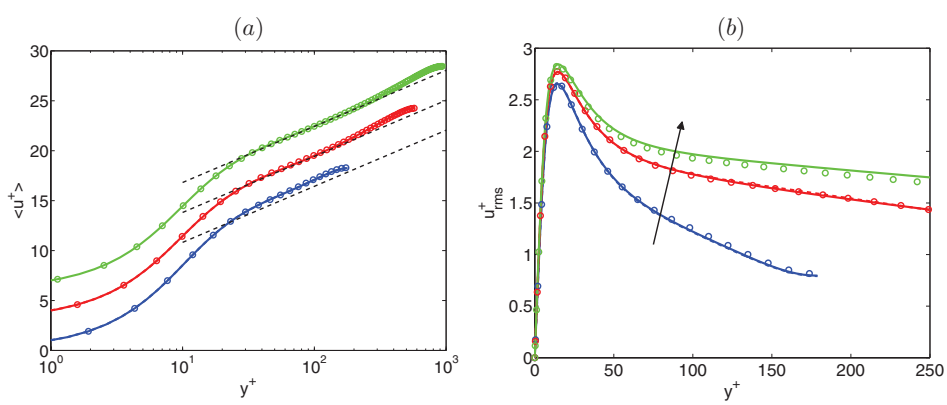

FIG. 1. (a) Mean streamwise velocity $\langle u \rangle$ at $Re_\tau = 180$, 590, 1000 with the $Re_\tau = 590$ simulations shifted 3 units along the y-axis, and the $Re_\tau = 1000$ simulation 6 units for clarity. The black dotted lines indicate the log law $1/\kappa \cdot \log(y^+) + B$ with $\kappa = 0.41$ and $B = 5.2$. (b) Fluctuations $u^+_{\rm rms}$. The full lines denote 180*c*, 590*m*, and 1000*m* simulations (see Table I), the dashed lines show the 180*m* and 590*f* simulations. (○): DNS data taken from Moser *et al.*[12] at $Re_\tau = 180$ and 590 and del Álamo *et al.*[13] at $Re_\tau = 950$. Throughout this article, a blue (black) line denotes the 180*c* simulation, a blue (black) dashed line the 180*m* simulation, a red (dark grey) line the 590*m* simulation, a red (dark grey) dashed line the 590*f* simulation, and a green (light grey) line denotes the 1000*m* simulation. The arrow indicates increasing Reynolds number.

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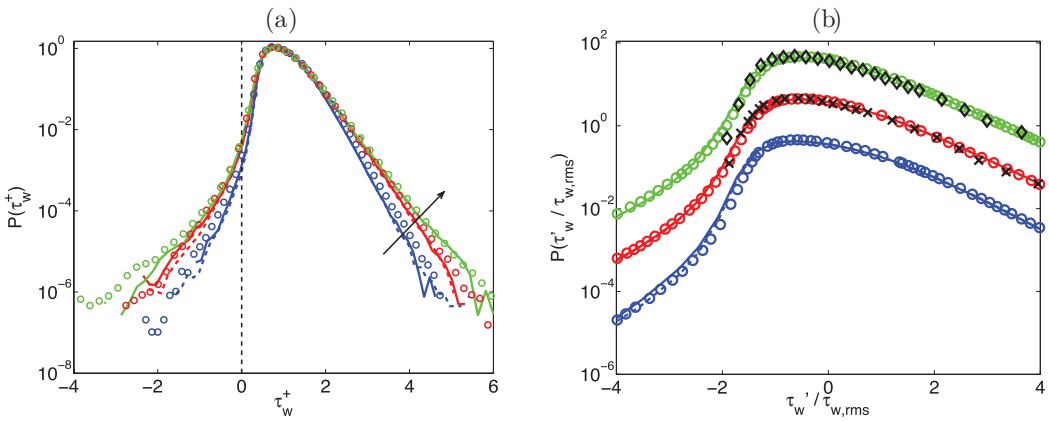

FIG. 2. Comparison of the PDF of τ_w of the present DNS with literature data. (a) Present results (lines) and (○) DNS of ZPG TBL (Ref. 14) at $Re_\tau = u_\tau \delta_{99}/\nu = 191$, 590, and 1000. The arrow indicates increasing Reynolds number. (b) Present results (lines) and (○) DNS data by Hu *et al.*[4] (at $Re_\tau = 180$, 720, and 1440, respectively), and experimental data by Grosse and Schröder[19] ((×): at $Re_\tau = u_\tau \delta_{99}/\nu = 3105$, (◊): at $Re_\tau = 4378$). For clarity, the red (dark grey) lines and dots, and the black crosses stem from results which are multiplied with a factor of 10; the green (light grey) lines and dots, and the black diamonds are multiplied with 100. The linecaption used is as in Figure 1.

III. RESULTS

A. Negative streamwise velocities

To investigate the existence of negative streamwise velocities, the PDF of the wall-shear stress $\tau_w = \mu \partial u/\partial y|_{\text{wall}}$, with $\mu = \nu\rho$ being the dynamic viscosity and ρ being the density, is studied, see Figure 2(a). For each simulation, N_f instantaneous velocity fields (see Table I) were taken to calculate the PDF of τ_w. Our results are compared with ZPG TBL data by Schlatter and Örlü[14] and Schlatter *et al.*[15] at similar Reynolds numbers. The PDFs show rare negative τ_w caused by negative streamwise velocities near the wall. We observe a Reynolds number dependence with the tails of the PDF widening for higher Reynolds number. Investigating the ensemble average of instantaneous flow fields as described in Sec. IV A shows that the typical shape of a region with backflow is circular with an average diameter at the wall of approximately 20 viscous units. This size proved to be independent of the Reynolds number, so the widening of the negative tail with increasing Reynolds number in the PDF can only be explained by a rise in the number of regions of backflow. Furthermore, the difference in the PDF of the 180*c* and 180*m* simulation is negligible, but the left tail of the 590*m* simulation is a bit wider than that of the 590*f*. The PDFs from the turbulent channel and the ZPG TBL collapse well onto each other when the wall-shear stress is scaled in viscous units, indicating that the existence of negative streamwise velocities near the wall and the physical phenomena causing it are universal for wall-bounded flows.

The third standardised central moment is called the skewness ($S(v) = \langle v'^3 \rangle / v_{\text{rms}}^3$) and the fourth standardised moment is called the flatness or kurtosis ($F(v) = \langle v'^4 \rangle / v_{\text{rms}}^4$). Table II shows the rms, skewness, and flatness values of the wall-shear stress for the simulations performed compared with experimental results for turbulent channel and boundary layer flow,[16] duct flow,[17] and DNS of ZPG TBL flow.[14] The results compare well for the different flow cases, with all three quantities increasing slowly with Reynolds number as also discussed by Örlü and Schlatter.[18]

Hu *et al.*[4] used DNS to calculate the wall-shear stress in a turbulent channel and found that the probability of negative wall-shear stress increases with Reynolds number which is consistent with our results. A comparison of the present results for the wall-shear stress fluctuations scaled with their rms value and DNS data by Hu *et al.*[4] is shown in Figure 2(b) where also data from micro-pillar experiments in a ZPG TBL flow by Grosse and Schröder[19] is added. The comparison is very good, confirming the general behaviour close to the wall. Note that the range of velocity measurements for the micro-pillar experiment is smaller, so they do not mention any negative wall-shear stress.

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TABLE II. RMS, skewness, and flatness of τ_w. The results stem from turbulent channel[16] ($Re_\tau = 200$), TBL (Ref. 16) ($Re_\tau = 1100$), and duct[17] measurements, and DNS of ZPG TBL flow.[14]

Re_τ	$\tau_{w\,\mathrm{rms}}^+$	$S(\tau_w)$	$F(\tau_w)$
Present results			
180*c*	0.367	0.930	4.257
180*m*	0.367	0.929	4.219
590*m*	0.407	1.019	4.802
590*f*	0.403	1.010	4.778
1000*m*	0.419	1.064	5.097
Alfredsson *et al.*[16]			
200	0.36	1.1	4.8
1100	0.39	1.0	4.8
Grosse and Schröder[17]			
900	0.36	0.95	4.47
Schlatter and Örlü[14]			
410	0.411	0.97	4.70
830	0.420	1.03	5.01
1230	0.426	1.07	5.19

To study how far away from the wall the negative velocity events extend into the flow, and how significant the fraction of negative velocities is as a function of wall distance y^+, the PDFs of the streamwise velocity u at selected y^+-values were also investigated. Such PDFs at $y^+ = 2$ and $y^+ = 5$ are depicted in Figure 3. The negative tail of the PDF is wider for higher Reynolds number at all y-positions indicating that negative velocities occur more often for higher Reynolds number, similar to the PDF of the wall-shear stress, see Figure 2(a). The difference between the 180 and the 590 simulations is larger than the difference between the 590 simulations and the 1000 simulation. Furthermore, backflow also occurs farther away from the wall for higher Reynolds number: the PDF at $y^+ = 5$ shows negative velocities for both the 590 simulations and the 1000*m* simulation, but no longer for the 180 simulations.

Summarising, negative velocities occur more often for higher Reynolds number as seen in Figure 4(a) which depicts the percentage of backflow as a function of the distance to the wall. Here, the percentage at every wall-normal position is calculated by taking the ratio of the area in the DNS with negative velocity and the total area in a wall-parallel plane. At $Re_\tau = 180$, approximately 0.01% of velocities are negative near the wall. This ratio rises to 0.05% at $Re_\tau = 590$ and to 0.06% at $Re_\tau = 1000$. They also occur further away from the wall for higher Reynolds number; up to $y^+ \approx 4.5$ at $Re_\tau = 180$, $y^+ \approx 6.5$ at $Re_\tau = 590$, and $y^+ \approx 8.5$ at $Re_\tau = 1000$. Small but consistent differences

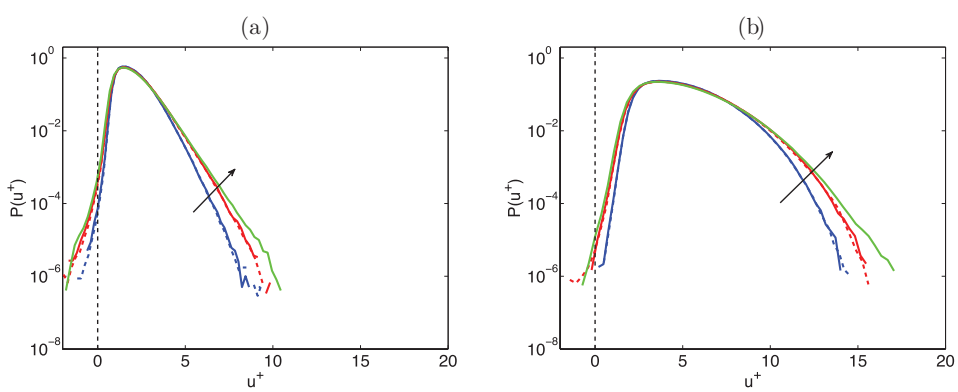

FIG. 3. PDF of the streamwise velocity u at (a) $y^+ = 2$ and (b) $y^+ = 5$. The linecaption used is as in Figure 1, with the arrow indicating increasing Reynolds number.

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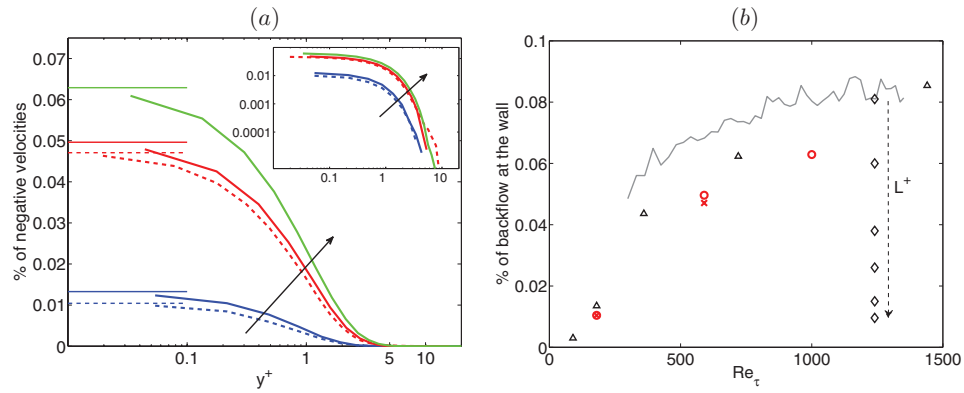

FIG. 4. (a) Percentage of negative velocities as a function of the distance to the wall. The horizontal lines indicate the percentage of negative wall-shear stress of the respective simulations. The linecaption used is as in Figure 1 with the arrow indicating increasing Reynolds number. The inset shows the same plot but with the y-axis in logarithmic scale. (b) Percentage of negative wall-shear stress. (○): 180c, 590m, and 1000m simulations; (×): 180m and 590f simulations. ZPG TBL results by Schlatter and Örlü[14] (solid line), and turbulent channel data by Hu *et al.*[4] (△). (◊): Backflow ratio for spatially averaged DNS data[14] to mimic a finite length of the sensing element, with $L^+ = [5.4, 21.7, 38.0, 54.3, 70.6, 87.0]$ in the direction of the arrow.

are observed between the simulations performed with different resolutions: higher resolution leads to slightly lower percentages of backflow. The inset in Figure 4(a) shows the same data, but with the y-axis scaled logarithmically. This scaling reveals the parallel behaviour of the area fraction at different Reynolds numbers, indicating a constant scaling factor. It also shows some scatter for the fine 590 simulation far away from the wall. Since the total number of backflow events here is only of order 1, the logarithmic plot becomes highly sensitive for small fluctuations.

Figure 4(b) shows the percentage of negative wall-shear stress as a function of Reynolds number. This percentage increases with Reynolds number, while the difference in results between resolutions is negligible. Both the DNS data from Hu *et al.*[4] and the TBL results of Schlatter and Örlü[14] show a similar trend although at slightly higher values. For the TBL results, we defined $Re_\tau = u_\tau \delta_{99}/\nu$ to compare turbulent channel and boundary layer results. Exemplary, the effect of insufficient spatial resolution, a common problem in near-wall measurements, is shown using the DNS data by Schlatter and Örlü[14] at $Re_\tau = 1240$. In order to mimic the finite length of a hot-wire or width of a hot-film sensor by means of the DNS data, the raw time series are filtered along the spanwise direction with a physical-space top-hat filter. There is a clear reduction of the percentage of backflow with increasing length of the sensing element, which goes along with observations on the fluctuation wall-shear stress by Örlü and Schlatter.[18] Hence, insufficient spatial (but also temporal) resolution might be one of the reasons why experimental investigations detect less or no instantaneous backflow.

B. High flatness and extreme wall-normal fluctuations

Near the wall, high flatness values for the wall-normal velocity are observed. The wall-limiting values of the flatness of the wall-normal velocity are shown in Figure 5(a). The flatness of v reaches approximately a value of 27 at $Re_\tau = 180$, 36 at $Re_\tau = 590$, and 43 at $Re_\tau = 1000m$. The error bars at $y^+ = 1$ indicate $\pm 2.5\%$ confidence intervals for the flatness as discussed in the Appendix. A clear Reynolds number dependence is seen with increasing flatness values for increasing Reynolds number, while the difference between resolutions is negligible. This contradicts preliminary results by Durst and Beronov[10] who observed a clear increase in flatness values for increasing resolutions. They attributed this increase to a numerical singularity near the wall, which introduces spurious error terms.

These high flatness factors are caused by very high fluctuations in the wall-normal velocities near the wall as discussed by Xu *et al.*[8] The PDF of v at $y^+ = 2$ for all simulations is shown in Figure 5(b). The very long tails in the PDF of v reveal the intermittent behaviour of the flow

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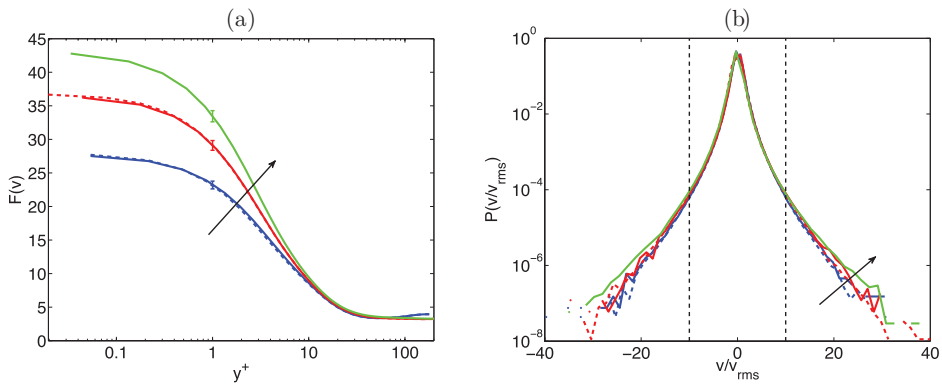

FIG. 5. (a) Flatness factor of the wall-normal fluctuations for the various simulations. The error bars at $y^+ = 1$ indicate $\pm 2.5\%$ confidence intervals. (b) PDF of the wall-normal velocity v at $y^+ = 2$ scaled with $v_{\rm rms}$. The dashed vertical lines designate extreme fluctuations $\pm 10 v_{\rm rms}$. The linecaption used is as in Figure 1, with the arrow indicating increasing Reynolds number.

field at this wall-normal position. The widening of the tails in Figure 5(b) is not as statistically significant as in Figure 3(a). This is because in Figure 5(b) the PDF is normalised with the rms of the wall-normal velocity which increases as the Reynolds number increases at a fixed wall-normal position. However, if the PDF of the wall-normal velocity is scaled in viscous units (not shown), the tail of the PDF clearly widens, just as in Figure 3(a).

Wall-normal velocity fluctuations larger than 10 times the local rms-value were defined extreme as indicated by the vertical dashed lines in Figure 5(b). These extreme fluctuations appear to be very rare and only occur in a region close to the wall as shown in Figure 6(a) where the percentage of the extreme v events is depicted as a function of wall-normal position. As with backflow, extreme wall-normal velocity fluctuations occur more often for higher Reynolds number, although the difference between Reynolds numbers is smaller than for the negative velocity events (from about 0.037% at $Re_\tau = 180$, up to 0.052% at $Re_\tau = 1000$). Analysis of the data shows that they do occur further away from the wall than the negative streamwise velocities, up to about $y^+ \approx 25$ for all simulations. This is confirmed by the inset, which shows the same plot, but with the y-axis scaled logarithmically. The Reynolds number dependence further away from the wall is fairly small and not as prominent as for backflow events. Also similar to the negative streamwise velocities, the simulations with higher resolutions show slightly lower percentages of extreme wall-normal velocity fluctuations.

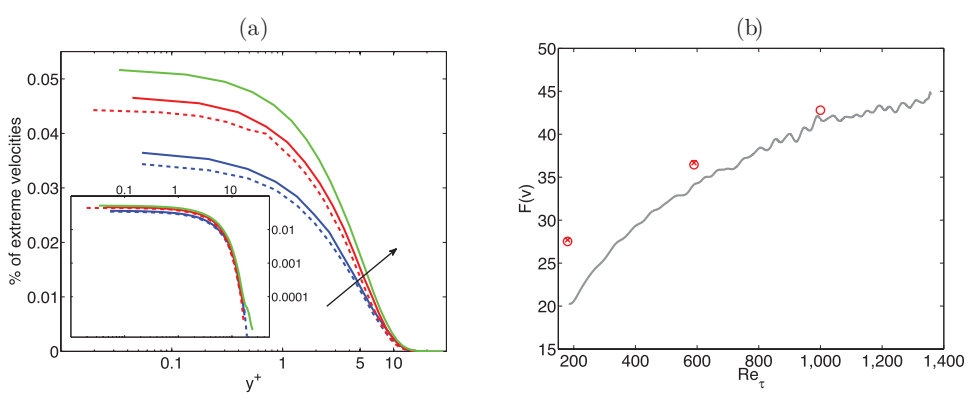

FIG. 6. (a) Percentage of extreme velocities as a function of the distance to the wall. The linecaption used is as in Figure 1, with the arrow indicating increasing Reynolds number. The inset shows the same plot but with the y-axis in logarithmic scale. (b) Flatness factor at the wall as a function of Reynolds number. (○): 180c, 590m, and 1000m simulations, (×): 180m and 590f simulations. Solid line: ZPG TBL results.[14]

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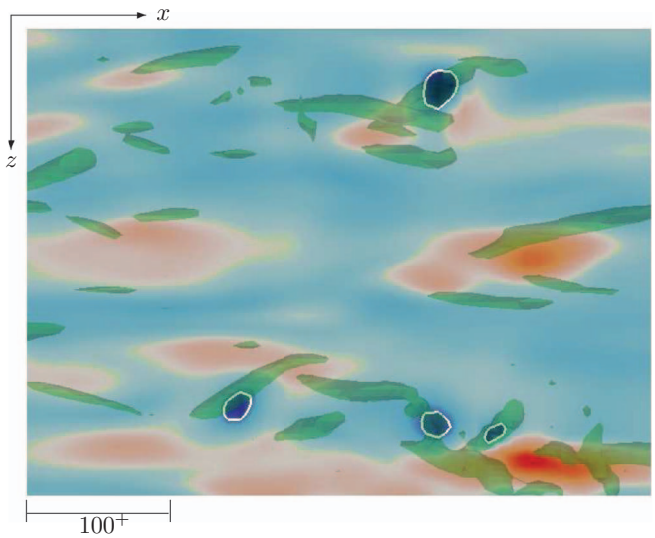

FIG. 7. Top view of part of the flow field of the 590*m* simulation, 500 × 400 viscous units in *x*- and *z*-direction, respectively. The background colour depicts the wall-shear stress with red (grey) representing high values and blue (dark grey) representing low values. The dark blue patches (black) encircled by white lines depict regions of negative τ_w. The green (dark grey) structures are isocontours of negative $\lambda_2^+ = -0.032$.

In Figure 6(b), the wall limiting value of the flatness of v is compared with ZPG TBL results by Schlatter and Örlü.[14] Their data shows similar results although with slightly lower values. Furthermore, the drop-off in flatness value for low Reynolds number is slightly faster than for our data. These TBL results disprove the claim by Lyons *et al.*[7] that the high flatness values are caused by the recycling of strong disturbances due to the use of periodic boundary conditions since TBL simulations do not use periodic boundary conditions. Also the results in Figure 5(a) refute their suggestion since the flatness at $Re_\tau = 1000$ is significantly higher than at $Re_\tau = 590$, while according to Lyons *et al.*[7] it should be lower since the boxsize used for the simulation at $Re_\tau = 1000$ is much larger and thus the effect of this recycling should be less pronounced.

IV. VISUALISATION

A. Negative streamwise velocities

Figure 7 shows part of a typical instantaneous streamwise velocity field of the 590*m* simulation. The background shows the wall-shear stress with high and low-speed streaks visible. Four regions of negative velocity are clearly visible and located in low-speed streaks.

To investigate the universality of these findings, an ensemble average of the velocity field around the regions with negative velocity is calculated. From all instantaneous fields, the ensemble averaged velocities and derivatives of the velocities in a box of size 75 × 45 × 75 viscous units in *x*-, *y*-, and *z*-direction, respectively, surrounding events with negative wall-shear stress are calculated. In order to compute a meaningful ensemble average, the orientation of the oblique vortex needs to be taken into account. This is done by mirroring the field in *z*-direction when necessary, ensuring that the vortex always has the same oblique orientation. After the ensemble averaged velocities are calculated, λ_2 is recomputed. Figure 8 shows the results of this spatial averaging over 292 events for the 590*f* simulation. Figure 8(a) clearly shows that the region of negative velocity is indeed located in a low-speed streak and that at the wall the region of negative velocity is approximately circular with an average diameter of approximately 20 viscous units. Furthermore, an inclined vortex lies on top of this region with an angle of about 45° in the *xz*-plane. Due to this inclination, the vortex induces flow in both (negative) spanwise and (negative) streamwise direction, i.e., transverse flow and backflow, respectively. This is further confirmed in Figure 8(b) where a *yz*-cut through the oblique vortex and the region with backflow is plotted. The centre of the vortex is clearly visible

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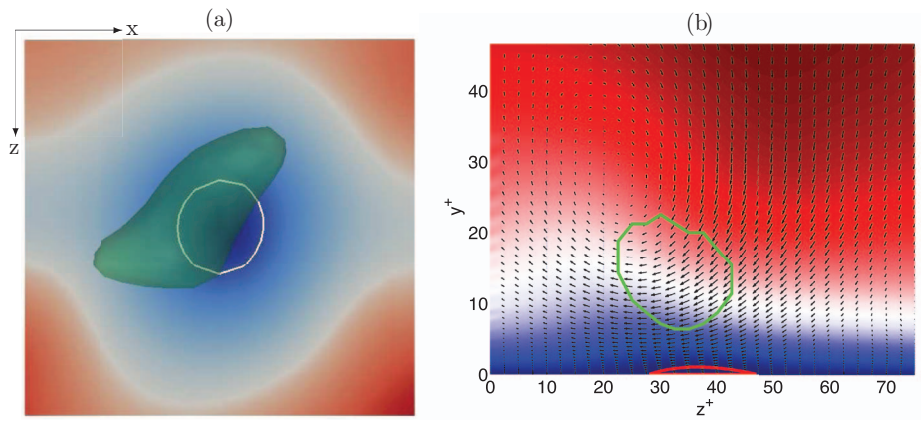

FIG. 8. (a) Top view of the ensemble averaged velocity field of negative velocity events, 75 × 75 viscous units large. The green (dark grey) structures are isocontours of negative $\lambda_2^+ = -0.032$. (b) The *yz*-plane cutting the region of negative velocity and the vortex as shown in Figure (a). The patch of negative *u* (within the red (grey) contour line) is induced by the oblique vortex on top, with background colour representing u^+. The green (dark grey) contour line represents the region where $\lambda_2^+ \leq -0.032$. In both (a) and (b) the same colour scheme as in Figure 7 is used.

at $(y^+, z^+) \approx (25, 22)$. Note that in this representation only the in-plane (spanwise) vortical motion can be identified. However, the vortex axis is oblique with respect to the *yz*-plane, so it also induces negative streamwise velocity near the wall. The ensemble average of negative streamwise events for the 590*m* was also calculated, but showed similar properties indicating that the resolution of the 590*m* simulation is sufficient to resolve all phenomena.

The generation of these oblique vortices is thought to be due to instability of streaks as discussed by Schoppa and Hussain.[20] The vortices causing backflow are vortices in the near-wall regeneration cycle that turn into extremely strong versions of the oblique vortices discussed by Schoppa and Hussain[20] and are therefore only observed rarely.

B. Extreme wall-normal velocity fluctuations

The extreme v events were previously studied by Xu *et al.*[8] who reported that the extreme events that contribute to the high flatness of the wall-normal velocity very close to the wall can be characterised by a positive and negative velocity spike (defined in Ref. 8 by $|v'/v_{\rm rms}| \geq 5$) which always appear in pairs. In addition, they noted that the dimension of each positive/negative spike is about 50 wall units in streamwise and 30 wall units in spanwise direction. These findings are confirmed in the present simulation although the spikes are found to be smaller, and similar events were also observed in the ZPG TBL simulations by Schlatter and Örlü.[14] Here, a typical example of both positive and negative wall-normal velocity spikes at $y^+ \approx 1$ taken from the 590*m* simulation is shown in Figure 9(a). Regions of extreme high/low wall-normal velocity fluctuations are designated by the white and black contour lines, respectively. In addition, vortices using the λ_2 criterion up to $y^+ \approx 25$ with a contour value of $\lambda_2^+ = -0.032$ are also shown in the same figure.

Once again, an ensemble average of the velocities in a box surrounding these extreme events was calculated to investigate the general properties. Figure 10(a) shows the spatially averaged velocity field over 178 events of the 590*f* simulation with extreme wall-normal velocity pairs at $y^+ \approx 1$ in a box of size 170 × 45 × 100 viscous units in *x*-, *y*-, and *z*-direction, respectively. It is seen that the spikes appear in pairs and each spike is elliptical orientated in streamwise direction. A spike has a typical streamwise length of about 25 and a spanwise width of about 7.5 viscous units. Both the width and length are less than those found by Xu *et al.*[8] Moreover, the negative wall-normal velocity spike is associated with a high-speed region indicating a strong sweep motion (downwash motion towards the wall), whereas the positive velocity spike is located somewhere between a high and low-speed streak. A quadrant analysis shows that sweeps with $v \leq -10v_{\rm rms}$ contribute about 5% to the total Reynolds shear stress at $y^+ \approx 1$ in the 590*m* simulation even though the area occupied by these

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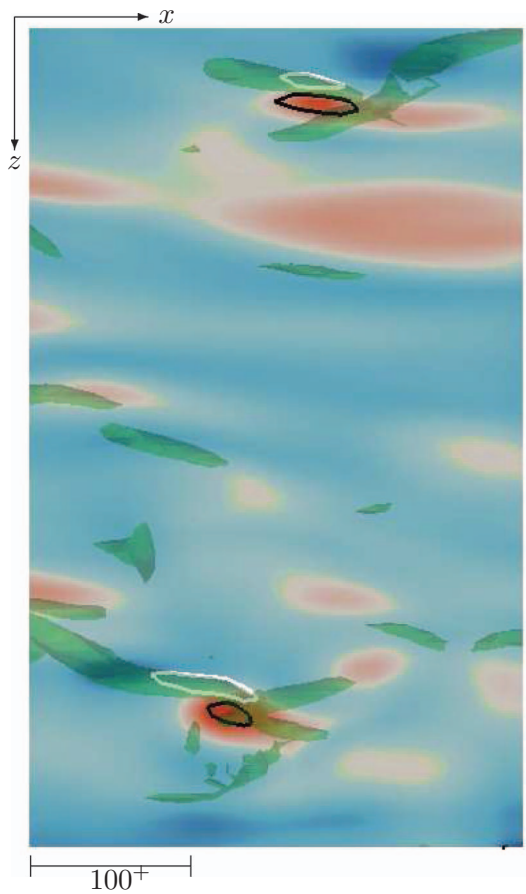

FIG. 9. Top view of part of the flow field of the 590*m* simulation at $y^+ \approx 1$, 300×500 viscous units in *x*- and *z*-direction, respectively. The background colour depicts *u*, the white and black contour lines represent regions with extreme positive and negative *v*, respectively. The green (dark grey) structures are isocontours of negative $\lambda_2^+ = -0.032$.

extreme sweeps is only 0.016% of the total area. This behaviour is consistent with the results found by Xu *et al.*[8] and the JPDF of $u/u_{\rm rms}$ and $v/v_{\rm rms}$ as shown in Figure 11. Here, extreme negative *v* velocities are correlated with high *u* values (i.e., sweeps), and extreme positive *v* values with medium *u* velocities. The ensemble average for the 590*m* simulations also confirms these properties for the pair of extreme wall-normal events.

Xu *et al.*[8] stated that the cause of the near-wall wall-normal velocity spikes is related to turbulence phenomena outside the viscous sublayer. According to these authors, the large negative spikes are results of the downwash motions coming from a near-wall vortex (being centred around $y^+ \approx 15$) which is a possible product of interaction of a vortex living in the buffer region (at about $y^+ \approx 30$) with the wall. However, no explanation is provided for the positive spike. In the present case, however, the phenomena appear to be different. There is always a quasi-streamwise vortex lying above the extreme events, see the *yz*-plane cut through the ensemble average of the pair of extreme *v* events as shown in Figure 10(b). The quasi-streamwise vortex with the centre at $(y^+, z^+) \approx (8, 45)$ gives rise to the upwash and downwash motions on its two sides which leads to both the positive and negative extreme *v* fluctuations.

C. Correlations

Both negative streamwise velocities and extreme wall-normal velocity fluctuations occur more often with higher Reynolds number. However, they do not appear to be correlated looking at their

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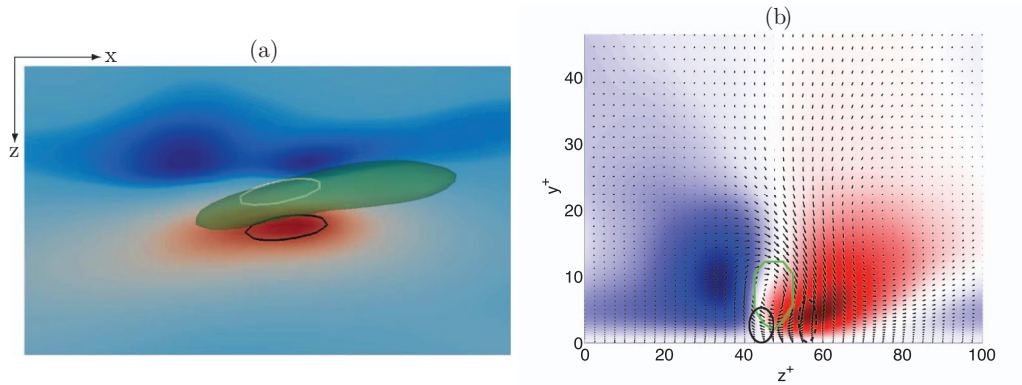

FIG. 10. (a) Top view of the ensemble averaged velocity field of extreme wall-normal velocity events, 170 × 100 viscous units large. The green (dark grey) structures are isocontours of negative $\lambda_2^+ = -0.032$. (b) The yz-plane cutting the extreme wall-normal velocity regions and the vortex as shown in Figure (a). The elliptic full contour line shows the extreme positive spike, the dashed contour the extreme negative spike. The green (dark grey) contour line represents the region where $\lambda_2^+ \leq -0.032$. In both (a) and (b) the same colour scheme as in Figure 9 is used.

JPDF as shown in Figure 11. Note that the contour lines are evaluated on a logarithmic scale to better visualise the rare events. Negative u events are equally distributed around $v/v_{\rm rms} = 0$ and do not occur more frequently for $|v/v_{\rm rms}| > 10$. As mentioned in Sec. IV B, extreme negative v velocities are correlated with high u values, while extreme positive v values are correlated with medium u velocities. So both positive and negative extreme wall-normal events are not correlated with negative streamwise velocity events. This lack of correlation was to be expected since backflow occurs predominantly in low-speed streaks, while extreme wall-normal velocity fluctuations occur in high-speed and in between streaks.

The JPDF of u and the pressure p', and v and p' shown in Figure 12 show interesting features. Figure 12(a) shows a correlation between negative u and negative p'. Note that negative p' is a necessary but not sufficient prerequisite for the presence of vortical motion. However, as seen in Figure 8, backflow is caused by a strong oblique vortex. So this correlation between negative u and negative p' corresponds to the oblique vortices. Furthermore, there appears to be a sharp cutoff line (indicated by the inclined dashed line with slope 20 running from extreme negative p', negative u events, towards extreme positive p', moderate u events). In Figure 12(b), extreme negative v events

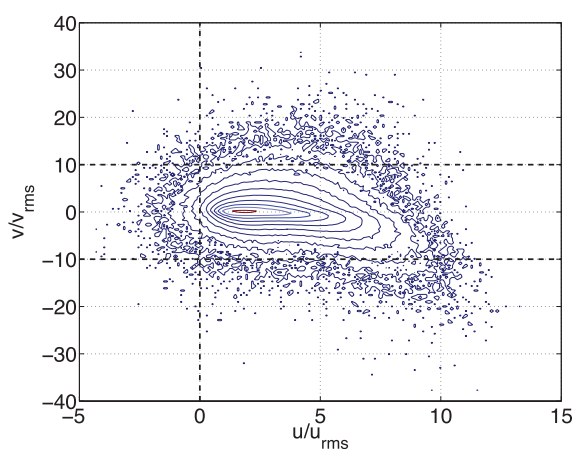

FIG. 11. The JPDF of u and v at $y^+ = 0.71$ for the 590m simulation. The 12 contour lines are logarithmically equally spaced between 10^{-6} and 10^0. The vertical dashed line indicates backflow, while the horizontal dashed lines indicate extreme fluctuations $\pm 10 v_{\rm rms}$.

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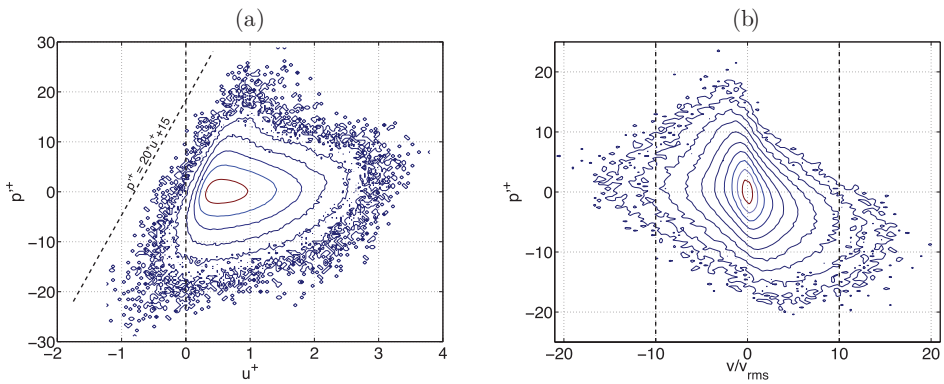

FIG. 12. The JPDF of (a) u and p', and of (b) v and p' at $y^+ = 0.71$ for the 590m simulation. In (a) the vertical dashed line indicates backflow with the inclined dashed line indicating a cutoff and in (b) the vertical dashed lines indicate extreme fluctuations $\pm 10 v_{\rm rms}$. The 12 contour lines are logarithmically equally spaced between 10^{-6} and 10^0.

are distributed symmetrically around $p' = 0$, while extreme positive v events are correlated with negative p'. Finally, positive p' is correlated with negative v events which indicate splatting events.

D. Outer layer influence

Since we observe a clear Reynolds number dependence for both types of rare events, and the regular turbulent structures in the viscous sublayer are assumed to be independent of Reynolds number, there must be an interaction between the rare events close to the wall and the Reynolds number dependent large-scale motion further away from the wall. Recent studies by Mathis *et al.*[21] and Schlatter and Örlü[22] among others discussed the modulation of small-scale structures near the wall by large-scale motions in the outer layer. Figure 13 shows the interaction between the large and small-scale motion for an instantaneous velocity field at $Re_\tau = 1000$, by comparing the position of locations where these rare events occur in relation with the large-scale motions in the streamwise velocity field at $y^+ \approx 100$. The locations with backflow were determined by points with negative wall-shear stress, and the position of extreme wall-normal events by points with extreme wall-normal velocity at $y^+ \approx 1.5$. To better visualise the large-scale motions, the velocity field was filtered with a Gaussian low-pass filter ($g(k_x, k_z) = 1 - \exp(-(k_x^2 + k_z^2)l/(2\pi)^2)$, with k_x and k_z being the wavenumber in streamwise and spanwise direction, respectively, and l being the filter width). The filter width was chosen to be larger than the size of the near-wall streaks, but smaller than the size of the large-scale structures.

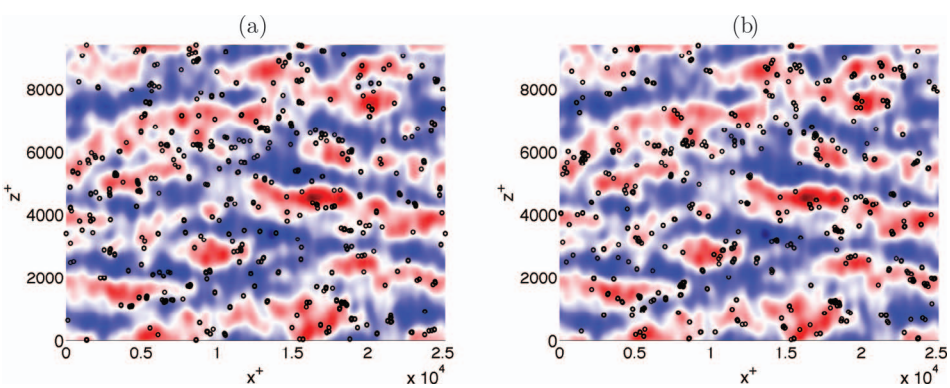

FIG. 13. The filtered streamwise velocity at $Re_\tau = 1000$ in an xz-plane at $y^+ \approx 100$ with the background colour representing u'^+ (red (dark grey): high speed and blue (grey): low speed). (a) (○): Points with negative wall-shear stress and (b) (○): points with extreme wall-normal velocity fluctuations at $y^+ \approx 1.5$.

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Most of the rare events occur below the large-scale motions with positive sign: 69% of the points with backflow and 77% of the points with extreme wall-normal velocity are located underneath a positive large-scale motion. High-speed streaks induce a higher intensity of fluctuations than low-speed streaks do.[22] This phenomenon is known as amplitude modulation. Due to this modulation, positive large-scale motions induce more intense velocity fluctuations than the negative large-scale motions. This explains why most of the extreme v-events are located underneath a positive large-scale motion. And even though the average u' is positive in a positive large-scale structure, the fluctuations are large enough to ensure that most of the backflow also occurs within positive large-scale motions. In negative large-scale motions, the fluctuations do not need to be as large to realise backflow, which explains the remaining 31% of points with negative wall-shear stress. The amplitude modulation effect becomes progressively stronger as the Reynolds number increases as noted by Mathis *et al.*[21] and Schlatter and Örlü.[22] This can explain why both kinds of rare events occur more often as the Reynolds number increases.

V. INFLUENCE OF MEASUREMENT ACCURACY

The present work is primarily based on DNS of channel flows, while its general character has been established for canonical wall-bounded flows as well. Consequently, the findings have direct consequences on state of the art measurement techniques applied in the sublayer of wall-bounded flows, which will be treated in the following. Starting from the early statement by Eckelmann[1] that "with certainty, there are no negative velocities near the wall," only few have detected[23] and reported[5] instantaneous backflow in their measurements. Also, the first DNS of a turbulent channel flow[24] showed considerably higher values of $F(v)$ as compared to available experimental data and in consequence stimulated several studies aimed at explaining the disparities between numerical and experimental results.[8–10]

A. Negative streamwise velocities

Classically, u within the viscous sublayer or its wall derivative, τ_w, has been measured through thermal anemometry, i.e., by means of hot-wire or hot-film probes[25,26] (see also the review paper by Hanratty and Campbell[27]) although MEMS based techniques[28,29] have also been used with varying success. These thermal methods are generally plagued by a number of problems,[30] such as the inherent forward-reverse ambiguity, which excludes the possibility to measure instantaneous reverse flow *per se*.[31] Hot-film arrays have been employed in the past, which in principle could detect backflow, but results based on these have shown "that there are no flow reversals at the wall."[2] Note, however, that the employed hot-films suffer from insufficient spatial resolution (cf. Ref. 18). Hot-film probes used in air utilising static calibrations, on the other hand, have been found to substantially attenuate the fluctuating magnitude.[16] While time resolution is generally not an issue for hot-wire probes, spatial resolution – in particular, for high Reynolds number measurements – constitutes a problem[32] as hinted upon in Figure 4(b), explaining most of the scatter observed in experimental studies concerning the magnitude of the wall-shear stress fluctuations.[18] Also, additional heat losses while approaching the wall seemingly increase the sensed mean, thereby decreasing the velocity sensitivity which reduces the amplitude of the measured fluctuations and consequently their ratio, the turbulence intensity.[33]

When it comes to the PDFs of u within the viscous sublayer or τ_w it has been common practice to compare the PDFs of the fluctuating wall-shear stress, independent of flow case and Reynolds number, scaled by its local standard deviation.[2,17,25,34,35] Such a plot is shown in Figure 14, depicting the same data as in Figure 2(b) with a general agreement between the data sets from different flow cases and Reynolds numbers. Also shown is a log-normal distribution (proposed by Alfredsson *et al.*[36]) which represents the compiled data sets satisfactorily well. The apparent good collapse of the data, often been highlighted in various studies based on such a plot, might be misleading: first of all, it does not allow direct identification of backflow; in fact, the log-normal distribution, which approximates the data well, is not defined for negatives values. Second, Reynolds number effects visible in the tails of the PDFs when shown on a logarithmic ordinate are not visible when

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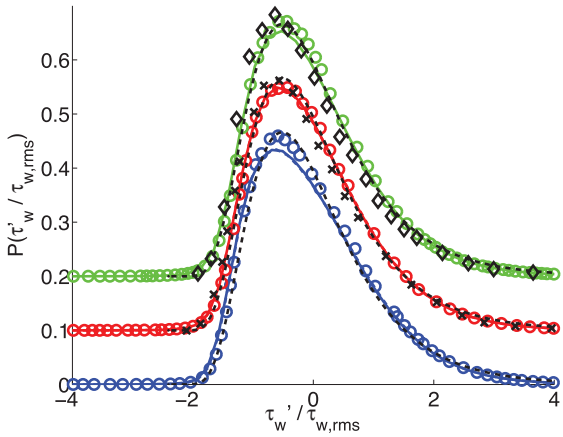

FIG. 14. Comparison of the PDF of τ_w of present DNS with literature data, with the same data as in Figure 2(b), shown in the commonly used lin-lin plot. The data is now shifted with 0.1 and 0.2, respectively, to clarify the picture. The dashed black lines represent a log-normal distribution as proposed in Ref. 36.

shown on a linear ordinate (compare Figures 2(b) and 14). Third, any attenuation of the fluctuating signal, be it through insufficient time or spatial resolution of the sensing element, will barely show up, since the standard deviation will similarly be reduced, thereby compensating the effect of attenuation on the PDF. Hence, a presentation of data as in Figure 14, albeit often done this way, is not recommended, despite its apparent collapse of available data irrespective of Reynolds number or quality of the experimental data. This is, in particular, of importance, when testing or validating innovative techniques that have in recent years been introduced to measure the fluctuating wall-shear stress.[19,35,37–39]

While the direct detection of backflow is in principal not possible by means of single hot-wire or hot-film probes, very few experimentalists have observed and/or documented the occurrence of backflow by means of LDV.[5,23] It should be kept in mind that special care needs to be taken when attempting to detect backflow experimentally and observe contiguous backflow events within the viscous sublayer when using LDV. First of all, the sensing volume needs to be located away from the wall, and the frequency of occurrence drastically reduces with distance from the wall (as apparent from Figure 4(a)). Second, in order to capture the strong velocity gradients within the viscous sublayer, the diameter of the LDV measurement volumes has to be as small as the viscous length scale,[40] which brings along a drastic reduction of the sampling rate (orders of magnitude smaller than those encountered in hot-wire or even hot-film measurements). Despite considerable longer sampling times, the practically achieved sample space is nonetheless limited, explaining partly the scatter commonly observed in higher-order statistics from LDV measurements.[41–43]

B. Extreme wall-normal velocities

When it comes to measurements of wall-normal velocities, LDV has been the most common technique within the sublayer,[44,45] while hot-wire probes have been used for the region beyond the viscous sublayer. X-wire probes have hereby successfully been used for near-wall measurements in general,[46,47] while only V-probes were able to provide reliable data within the sublayer.[48] Since the controversy on the high wall-normal flatness values has mainly been driven by DNS and LDV results, we will restrict the discussion here on LDV-related problems. Some of these have already been mentioned in Sec. V A, such as spatial and temporal resolution as well as the low data rates and limited sample spaces.

Another reason for the low $F(v)$ values in the LDV measurements is the smoothing operation used in the post-processing of LDV data.[7] A more common practice, however, is the removal of seemingly unphysical data points through "dropout," which practically means to omit data points

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exceeding a certain user-defined multiple of the local standard deviation within a moving window. Values as low as ±2.75,[49] 6,[50] and 7,[41] and up to 10 (Ref. 42) standard deviations have been reported in the literature. However, it was shown by Xu *et al.*,[8] that even a filter accepting data within 10 local standard deviations would considerably reduce the value of $F(v)$ within the viscous sublayer due to the intermittent nature of the signal. Note, in this respect, that in the present study, extreme events are defined as those excursions exceeding 10 standard deviations; i.e., the present study considers events which are commonly considered as "unphysical" in LDV and are usually removed. It is interesting to note that the application of the aforementioned filter was "found to eliminate any spurious data without changing the shape of the probability density distributions";[41] emphasising again the note of caution issued regarding the utilisation of the PDF as depicted in Figure 14. Furthermore, it is also interesting to note that, e.g., the "number of outliers found per time series" in the experiments by den Toonder and Nieuwstadt[42] "of about 18 000 samples was always less than 3, and they occurred only in the time series measured at positions less than 0.5 mm from the wall," which nicely agrees with the results presented in Figure 6(a).

Coming back to the common feature of backflow and strong v events, it is obvious that the measurement technique used has to be tuned in order to detect such events ("previous experiments were in general not designed to observe such spikes"[8]) and that a great portion of luck and patience is required to detect them, in particular, in single-point measurements, so, e.g., Westerweel and Boersma[51] report that only after "a total measurement time of 6 h such an event was detected in the unfiltered LDA time series that strongly resembled such rare events found in a DNS."

VI. CONCLUSIONS

DNS data of turbulent channel flow at three different friction Reynolds numbers $Re_\tau = 180, 590$, and 1000 has been used to investigate rare events near the wall, i.e., negative streamwise velocities and extreme wall-normal velocity fluctuations. A grid refinement is performed to also investigate potential resolution effects. Negative streamwise velocity is indeed found in the flow field in the vicinity of the wall (i.e., negative wall-shear stress) and this flow reversal is not due to numerical artifacts. With an increase of Reynolds number, the percentage of flow reversal increases from 0.01% at $Re_\tau = 180$ to 0.06% at $Re_\tau = 1000$. These results are consistent with previous studies by Hu *et al.*[4] Backflow also extends further away from the wall with increasing Reynolds number, up to $y^+ \approx 4.5$ for $Re_\tau = 180$ and $y^+ \approx 8.5$ for $Re_\tau = 1000$. The simulations performed at a higher resolution suggest a slightly lower percentage of backflow, but neither qualitative nor significant quantitative differences are found suggesting that the resolution is sufficient. These regions where backflow occurs are typically circular with an average diameter based on ensemble averages, of approximately 20 viscous units at the wall, and the backflow is induced by strong oblique vortices that are found outside the viscous sublayer on top of these regions. These oblique vortices also induce transverse flow.

Near the wall, extreme wall-normal velocity fluctuations are also observed. These extreme fluctuations cause high values of the flatness near the wall of up to $F(v) = 43$ for $Re_\tau = 1000$. The flatness values are increasing with increasing Reynolds number and the resolution effects on the wall-limiting values of flatness are negligible. Similar to the negative u events, the extreme v events occur more frequently for higher Reynolds number, from 0.037% at $Re_\tau = 180$ to 0.052% at $Re_\tau = 1000$. In addition, changing the resolution has similar effects, i.e., with higher resolution, the percentage of extreme high wall-normal velocity fluctuations slightly decreases. Positive and negative velocity spikes appear in pairs, and are located on the two sides of a strong streamwise vortex. The spikes are elliptical and orientated in streamwise direction with a typical length based on ensemble averages of 25 and a typical width of 7.5 viscous units at $y^+ \approx 1$. The negative spike occurs in a high-speed streak indicating a strong sweeping motion, while the positive spike is located between a high and low-speed streak. The present results are consistent with previous studies by Xu *et al.*[8] at $Re_\tau = 180$.

Since the negative streamwise velocities and extreme wall-normal velocity fluctuations occur in different regions in relation to high and low-speed streaks, it can be concluded that they are uncorrelated. This is also confirmed by investigating the JPDF of u and v which shows no correlation between

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negative u and extreme v events. Most of the extreme events (69% of backflow events and 77% of extreme v events) occur underneath positive large-scale motions which are the more active ones and thus allow for higher velocity fluctuations. This can explain the Reynolds number dependence since these large-scale motions become more pronounced with increasing Reynolds number.

The similarity of the results found in experiments and DNS of turbulent boundary layers and turbulent channels for both the negative streamwise as well as the extreme wall-normal fluctuations suggest these events are universal for wall-bounded flows. In order to detect both these rare events in experiments, measurement techniques have to be specifically tuned. Special care has to be taken to ensure that both the spatial and temporal resolution is sufficient to register these rare events.

ACKNOWLEDGMENTS

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APPENDIX: STATISTICAL CONVERGENCE OF THE DATA

In this Appendix we discuss the results from our study on the statistical convergence of the data. We introduce two new quantities: N_p and N_p^*. N_p is the number of independent data points (measurements) used to calculate the statistics. To determine N_p it is assumed that the integral length scale in the spanwise direction is on the order of the mean streak spacing which is about 100 viscous units. The integral length scale in the streamwise direction is about 300 viscous units determined from the corresponding autocorrelation. With these estimates the number of independent points in each wall-normal plane for the simulations can be calculated, yielding 49, 216, and 7802 independent points in each wall-normal plane for the simulations at $Re_\tau = 180$, 590, and 1000, respectively (see Table III). Note that this number of independent points per wall-normal plane depends on the boxsize and Reynolds number, but not on the resolution.

The second new quantity N_p^* represents the minimal number of points necessary to ensure statistical convergence of the data. Figure 15 shows the evolution of the flatness for all velocity components $F(u)$, $F(v)$, and $F(w)$ for the 180c simulation at $y^+ = 1$ as a function of the number of independent points N_p used. In general, all three flatnesses show an initial transient phase, i.e., large fluctuations are clearly visible, due to the insufficient number of independent points that are used to calculate the average. If sufficient independent points are used for averaging (here $N_p^* = 5145$ as indicated by the vertical dashed line, which corresponds to 105 instantaneous velocity fields), the curves stay within a certain bounded interval. The intervals indicated by the horizontal dashed lines represent $\pm 2.5\%$ intervals around the mean of the respective variables, with the mean calculated starting from the point of convergence. It is seen that $F(u)$ and $F(w)$ converge comparably fast, while $F(v)$ takes a bit longer to converge. This indicates that the fluctuations in the wall-normal velocity component are more intermittent than in the other two components. The evolution of these variables

TABLE III. N_p the number of independent points and N_p^* the number of independent points necessary to ensure statistical convergence.

Re_τ	N_p per wall-parallel plane	N_p^*
180c	49	5145
180m	49	2499
590m	216	5832
590f	216	2592
1000m	7802	7802

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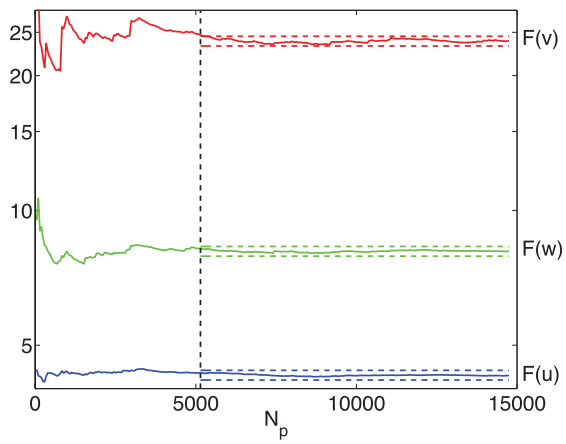

FIG. 15. Evolution of $F(u)$ (blue (black)), $F(v)$ (red (dark grey)), and $F(w)$ (green (light grey)) as a function of the number of independent points N_p used to calculate these quantities for the 180c simulation at $y^+ = 1$. The dashed vertical line at $N_p = 5145$ denotes N_p^*, the point from which the data is assumed to be sufficiently converged. The dashed horizontal lines represent the $\pm 2.5\%$ confidence intervals for the respective variables.

for the other simulations shows similar results with a critical N_p^* value of the same order as shown in Table III. Note that for the simulation at $Re_\tau = 1000$ $N_p = N_p^*$ since for the boxsize used, the data of one realisation of a velocity field is sufficient to ensure convergence.

We can conclude that the number of independent samples should be of the order of $N_p^* = 5000$ to have well converged statistics. This condition is satisfied by all our DNSs. This required number of independent samples corresponds well with results by Klewicki and Falco[52] who in their turbulent boundary study found that to ensure that the flatness is converged and stays within a $\pm 5\%$ interval around the mean, 2500 independent samples needed to be taken. Since our interval is chosen to be slightly more restrictive, it is logical that slightly more points are required to ensure convergence. Furthermore, the results by Klewicki and Falco[52] show that to ensure convergence of lower order moments less independent measurements are necessary which is confirmed (not shown) by our study, so also convergence of the mean velocities, rms-values and skewness is ensured.

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